

Task 'Unit 4: Cloud Native Technology Part 1'

Formative Activity Bash Script and Report - Automating Cloud Operations Using Scripting

The use of software and scripts to automate cloud operations such as infrastructure provisioning, deployment and tasks with limited manual intervention. In this document, Bash and the Azure Command-Line Interface (Azure CLI) are utilized to provision and configure an Azure virtual machine (VM). Microsoft describes Azure CLI as a way to build virtual machines and associated resources through commands, making deployment repeatable.

(Microsoft, 2026a).

The script was first developed and intended to create a resource group and then provisions a Linux Ubuntu VM. It executes important configurations such as the VM name, location, operating-system image, size and administrator account. The `--generate-ssh-keys` option provides a cryptographic secure authentication login method, without requiring a visible password that needs be written into the script. After the creation is done, the `az vm run-command` invoke executes commands inside the VM. The script updates the Ubuntu system, installs Nginx, enables the service and presents the hostname and system uptime. Microsoft documents Run Command as a method for executing shell scripts inside Azure Linux VMs.

(Microsoft, 2026b).

One important benefit of using scripting is repeatability. One script, or A template could be used repeatedly to re-produce identical settings multiple times, while executing changes with minimal, manual administrative effort and task repetition.

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This help to reduce deviation between workloads deployment and from the desired state, what makes deployments more effective. Infrastructure-as-code techniques are used to treat infrastructure configuration uniformly, as code, enabling resources to be constructed and replicated systematically.

(NIST, 2024).

Using scripts helps to improve speed and efficiency. Manual provisioning requires much administrative effort, for instance executing commands and configuration tasks. The use of automation helps to achieve these steps in a predictable, predefined series of sequences. AWS recommends automating operational activities as the use of automation can help to reduce manual effort, improve consistency and reliability of results, in addition to supporting improved human resources capacities.

(Amazon Web Services, 2024).

The use of automation also improves scalability. After scripts have been tested and validated, the variables can be adjusted and replicated to provision additional resources on additional environments. This brings additional benefit for organizations that requires consistency across configurations and environments such as development, testing and production. Academic research also shows that cloud automation can support dynamic resource provisioning and optimization, although automated environments still require monitoring and cost controls.

(Yang et al., 2024).

in summary, automation should complement human oversight, rather of replacing it, Scripts should be carefully crafted by experienced staff and tested before production

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use, credentials must be protected, permissions must follow least-privilege principles and unnecessary resources should be removed to improve simplicity. Overall, scripting provides a simple and practical method for improving cloud operations and making it faster, repeatable, consistent and easier to manage.

References

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